SAARATHI SMART AI TEACHER

An AI-Powered Interactive Learning Platform

Submitted for:

AI for Bharat Innovation Challenge/ Hackathon

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Introduction :

Education is increasingly moving toward digital platforms, but many existing online learning systems still rely on static content such as recorded videos and PDFs. These platforms often fail to adapt to the individual learning speed and understanding level of students, making it difficult for learners to grasp complex concepts. To address this issue, Saarathi — Smart AI Teacher is developed as an AI-powered tutoring platform that provides interactive and personalized learning support. The system acts as a virtual teacher that answers student questions, explains concepts step-by-step, and generates supporting materials such as diagrams and quizzes. Students can interact with the platform using text or voice input, and the AI generates explanations according to the learner's skill level. The response is delivered through a digital avatar with voice output, creating an engaging and interactive learning experience.

Steps of the Proposed System :

User Login:The student logs into the platform to access the AI teacher.

Question Input:The user asks a question through text or voice input.

Query Processing:The system sends the query to the AI model for analysis.

Content Generation:The AI generates explanations, diagrams, and learning materials.

Voice and Avatar Output:The response is converted into speech and delivered through the digital avatar.

Quiz Generation:The system creates quizzes to test the student's understanding.

Progress Tracking:The platform stores user activity and quiz scores to track learning progress.

Workflow :

1. User Interaction
The user interacts with the Saarathi platform through the web interface.

2. Question Input
The student enters a question using text or voice input.

3. Request Processing
The frontend sends the request to the backend server through an API.

4. AI Analysis
The backend forwards the query to the AI model for understanding and processing.

5. Content Generation
The AI generates explanations and may also create diagrams, images, or quizzes.

6. Speech Conversion
The response is converted into speech using text-to-speech technology.

7. Avatar Presentation
The digital avatar presents the explanation with synchronized voice output.

8. Response Display
The final output is displayed on the user interface.

9. Data Storage
The system stores user questions, quiz results, and learning activity in the database.

10. Progress Tracking
The stored data helps track learning progress and improve future recommendations.

Implementation :

1. System Architecture
Saarathi is implemented as a full-stack web application.

2. Frontend Development
The frontend is built using React and Vite for a fast and responsive user interface.

3. User Interaction
Students can interact with the AI teacher through text or voice input.

4. Learning Interface
The interface displays explanations, diagrams, quizzes, and learning progress.

5. Avatar Integration
An animated digital avatar presents responses visually to make learning more engaging.

6. Backend Development
The backend is developed using FastAPI to manage API requests and system operations.

7. AI Processing
The backend sends user queries to AI models and receives generated explanations.

8. Database Management
Supabase PostgreSQL stores user data, questions, quiz scores, and learning history.

9. Cloud Integration
Cloud services are used to support AI processing, speech generation, and media storage.

10. Scalable System
The system is designed to provide efficient and real-time interactive learning support.

11. Tools Used: React, Vite, FastAPI, Supabase (PostgreSQL), AI Models, Web Speech API, Text-to-Speech, AWS Cloud Services.

Conclusion :

1. Saarathi is an AI-based learning platform.
2. It provides interactive and personalized education.
3. Students can ask questions using text or voice.
4. The system gives explanations, diagrams, and quizzes.
5. It tracks student learning progress.
6. Saarathi makes learning easier and more effective.